

Jiadong Liang

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ACADEMIC POSITION

University of Pennsylvania
Postdoctoral researcher

Aug. 2024 -
Advisor: Prof. Yuxin Chen

EDUCATION

Peking University
Ph.D. of Statistics in School of Mathematical Sciences
Peking University
Bachelor of Statistics in School of Mathematical Sciences

Sep. 2019 - Jul. 2024
Advisor: Prof. Zhihua Zhang
Sep. 2015 - Jun. 2019

RESEARCH INTERESTS

Stochastic Optimization

- Design provably efficient online stochastic algorithms for (non)-convex optimization problems with various types of random oracles.

Computational Efficient Statistics

- Discover techniques for efficiently handling statistical estimation and uncertainty quantification tasks with minimal computational overhead.

Generative Model Theory

- Analyze the relationship between the performance of generative models (such as diffusion models) and their sample complexity and training methods.

PUBLICATIONS

* indicates equal contribution or alphabetical listing.

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| Decoupled Functional Central Limit Theorems for Two-Time-Scale Stochastic Approximation
Yuze Han*, Xiang Li*, Jiadong Liang* , Zhihua Zhang*
<i>Accepted by Mathematics of Operations Research (MOR)</i> | 2026 |
| Stochastic Approximation MCMC, Online Inference, and Applications in Optimization of Queueing Systems
Xiang Li*, Jiadong Liang* , Xinyun Chen, Zhihua Zhang
<i>Operations Research (OR)</i> | 2026 |
| Estimation and Inference in Distributional Reinforcement Learning
Liangyu Zhang, Yang Peng, Jiadong Liang , Zhihua Zhang
<i>Annals of Statistics (AoS)</i> | 2025 |
| Low-dimensional adaptation of diffusion models: Convergence in total variation
Jiadong Liang , Zhihan Huang, Yuxin Chen
<i>Partially accepted at Conference on Learning Theory (COLT); revise & resubmit at Journal of the American Statistical Association (JASA)</i> | 2025 |
| A Statistical Analysis of Polyak-Ruppert Averaged Q-learning
Xiang Li, Wenhao Yang, Jiadong Liang , Zhihua Zhang, Michel I. Jordan
<i>International Conference on Artificial Intelligence and Statistics (AISTATS)</i> | 2023 |

Asymptotic Behaviors of Projected Stochastic Approximation: A Jump Diffusion Perspective Jiadong Liang, Yuze Han, Xiang Li, Zhihua Zhang <i>Neural Information Processing Systems (NeurIPS), Oral presentation (top 1.5%)</i>	2022
Statistical Estimation and Inference via Local SGD Xiang Li, Jiadong Liang, Xiangyu Chang, Zhihua Zhang <i>Conference on Learning Theory (COLT)</i>	2022
Lipschitz Generative Adversarial Nets Zhiming Zhou, Jiadong Liang, Yuxuan Song, Lantao Yu, Hongwei Wang, Weinan Zhang, Yong Yu, Zhihua Zhang <i>International Conference on Machine Learning (ICML)</i>	2019

PREPRINTS AND WORKING PAPERS

* indicates equal contribution or alphabetical listing.

Optimal Training-Conditional Regret for Online Conformal Prediction Jiadong Liang, Zhimei Ren, Yuxin Chen <i>Submitted to Annals of Statistics (AoS)</i>	2026
Gradient Tracking for High-Dimensional Federated Optimization Jiadong Liang, Yang Peng, Zhihua Zhang <i>Preparation for submission</i>	2023
Online Statistical Inference for Nonlinear Stochastic Approximation with Markovian Data Xiang Li, Jiadong Liang, Zhihua Zhang <i>Submitted to Journal of Machine Learning Research (JMLR)</i>	2023
Asymptotic Behaviors and Phase Transitions in Projected Stochastic Approximation: a Jump Diffusion Approach Jiadong Liang, Yuze Han, Xiang Li, Zhihua Zhang <i>Submitted to Journal of Machine Learning Research (JMLR)</i>	2023

PRESENTATIONS AND TALKS

Stochastic Approximation MCMC And Online Inference For Dynamic Pricing And Capacity Sizing In Queues INFORMS Annual Meeting, <i>Phoenix, AZ, USA</i>	Oct 2023
Asymptotic Behaviors of Projected Stochastic Approximation: A Jump Diffusion Perspective NeurIPS Deep-dive Paper Panels, <i>Virtual</i>	Dec 2022
Optimal Training-Conditional Regret for Online Conformal Prediction International Seminar on Selective Inference, <i>Virtual</i>	Mar 2026

PROFESSIONAL SERVICES

Journal Reviewer for:

- Journal of Machine Learning Research
- Mathematics of Operations Research
- SIAM Journal on Mathematics of Data Science

Conference Reviewer for:

- ICML [2023], Neurips [2022, 2023, 2024, 2025], ICLR [2022], UAI [2020, 2025]

SKILLS

Languages: Chinese, English

Coding: Python, $\text{\LaTeX}$